

Olympiad Test Paper — Mathematics (Class 7)

Chapter 11: Finding Common Ground

Subject	Mathematics (NCERT Class 7)
Chapter	11 — Finding Common Ground (HCF & LCM)
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Decide first whether the problem asks for the largest number that DIVIDES the given numbers (HCF — for sharing/cutting/tiling) or the smallest number they DIVIDE INTO (LCM — for bells/lights/cycles meeting again). Beware the classic traps: HCF is never bigger than the smaller number, LCM equals the product only for co-prime pairs, and for the HCF you take the LOWEST powers of the common primes while the LCM takes the HIGHEST powers of all primes. Do not write on this paper — mark answers on your answer sheet.

1. The HCF of two numbers is the:

- (A) smallest number that both numbers divide into
- (B) largest number that divides both numbers exactly
- (C) product of the two numbers
- (D) sum of all common factors of the two numbers

2. Find the HCF of 24 and 36.

- (A) 6
- (B) 4
- (C) 12
- (D) 72

3. Find the LCM of 12 and 18.

- (A) 36
- (B) 6
- (C) 72
- (D) 216

4. Sameeksha tiles a 12 ft × 16 ft floor with the largest equal square tiles of whole-foot side that fit exactly. The side of each tile is:

- (A) 16 ft
- (B) 2 ft
- (C) 4 ft
- (D) 48 ft

5. For the same 12 ft × 16 ft floor, the number of largest square tiles needed is:

- (A) 6
(C) 48
- (B) 4
(D) 12

6. Which of these pairs is co-prime (HCF = 1)?

- (A) 8 and 15
(C) 8 and 12
- (B) 9 and 15
(D) 10 and 25

7. The LCM of two co-prime numbers 7 and 9 is:

- (A) 1
(C) 63
- (B) 16
(D) 7

8. Two bells ring at intervals of 6 minutes and 8 minutes. If they ring together now, after how many minutes do they next ring together?

- (A) 48
(C) 2
- (B) 14
(D) 24

9. The product of two numbers is 216 and their HCF is 6. Their LCM is:

- (A) 36
(C) 30
- (B) 1296
(D) 210

10. For any two numbers, $\text{HCF} \times \text{LCM}$ equals:

- (A) the sum of the two numbers
(C) the larger of the two numbers
- (B) the product of the two numbers
(D) the difference of the two numbers

11. Using $12 = 2^2 \times 3$ and $18 = 2 \times 3^2$, the HCF of 12 and 18 is:

- (A) $2^2 \times 3^2 = 36$
(C) $2^3 \times 3 = 24$
- (B) $2^2 \times 3 = 12$
(D) $2 \times 3 = 6$

12. Using $12 = 2^2 \times 3$ and $18 = 2 \times 3^2$, the LCM of 12 and 18 is:

- (A) $2 \times 3 = 6$
(C) $2^2 \times 3^2 = 36$
- (B) $2^2 \times 3 = 12$
(D) $2^3 \times 3^3 = 216$

13. Two numbers have HCF 6 and LCM 72. If one number is 24, the other is:

- (A) 36
(C) 12
- (B) 18
(D) 8

14. The longest tape that can measure 7 m, 3.85 m and 12.95 m an exact whole number of times is:

- (A) 7 cm
(C) 35 cm
- (B) 5 cm
(D) 70 cm

15. Which statement is always TRUE?

- (A) The HCF of two numbers is never larger than the smaller number.
(B) The LCM of two numbers is never larger than the larger number.

(C) The HCF of two numbers is always one of the two numbers.

(D) The LCM of two numbers always equals their product.

16. Find the HCF of 18, 24 and 30.

(A) 2

(B) 3

(C) 12

(D) 6

17. Find the LCM of 4, 6 and 8.

(A) 24

(B) 12

(C) 48

(D) 16

18. Three traffic lights change every 24 s, 36 s and 54 s. Having just changed together, after how long do all three change together again?

(A) 108 s

(B) 216 s

(C) 6 s

(D) 324 s

19. The greatest number that divides both 70 and 125 leaving remainder 0 is:

(A) 25

(B) 7

(C) 5

(D) 10

20. Two numbers are in the ratio 3 : 4 and their HCF is 5. The two numbers are:

(A) 3 and 4

(B) 9 and 12

(C) 30 and 40

(D) 15 and 20

21. The LCM of 15 and 20 is 60. Their HCF is:

(A) 5

(B) 3

(C) 4

(D) 300

22. Which of the following is NOT a common multiple of 4 and 6?

(A) 12

(B) 18

(C) 36

(D) 24

23. The number of common factors of 24 and 36 is:

(A) 3

(B) 4

(C) 6

(D) 12

24. Two gears have 8 and 12 teeth and mesh together. After how many teeth of the larger gear pass do they return to their starting position together?

(A) 96 teeth

(B) 2 teeth

(C) 4 teeth

(D) 24 teeth

25. The HCF of two numbers is 1. Which must be true?

(A) The numbers are co-prime and their LCM equals their product.

(B) Both numbers are prime.

(C) The numbers are equal.

(D) One number divides the other.

26. The HCF of 1794 and 2346 by repeated subtraction/division (Euclid) is:

- (A) 23 (B) 138
(C) 46 (D) 6

27. The smallest number that is exactly divisible by each of 1, 2, 3, 4, 5 and 6 is:

- (A) 30 (B) 720
(C) 120 (D) 60

28. Two ribbons of length 36 cm and 48 cm are cut into equal pieces of the greatest possible length with none left over. The length of each piece is:

- (A) 6 cm (B) 12 cm
(C) 4 cm (D) 24 cm

29. For the ribbons in the previous question, the total number of equal pieces obtained is:

- (A) 12 (B) 4
(C) 7 (D) 8

30. If two numbers are both multiples of their HCF h , then h is also a factor of:

- (A) their sum and their difference (B) only the larger number
(C) neither their sum nor their difference (D) only their product, never their sum

31. The prime factorisation of 360 is:

- (A) $2^3 \times 3 \times 5$ (B) $2^2 \times 3^2 \times 5$
(C) $2^3 \times 3^2 \times 5$ (D) $2^3 \times 3^2 \times 5^2$

32. The HCF of $2^2 \times 3 \times 5$ and $2 \times 3^2 \times 7$ is:

- (A) $2^2 \times 3^2 \times 5 \times 7 = 1260$ (B) $2^2 \times 3^2 = 36$
(C) $2 \times 3 \times 5 \times 7 = 210$ (D) $2 \times 3 = 6$

33. The LCM of $2^2 \times 3 \times 5$ and $2 \times 3^2 \times 7$ is:

- (A) $2 \times 3 = 6$ (B) $2^2 \times 3^2 \times 5 \times 7 = 1260$
(C) $2^3 \times 3^3 \times 5 \times 7$ (D) $2^2 \times 3 \times 5 = 60$

34. The greatest number that divides 43, 91 and 183 leaving the same remainder 3 in each case is:

- (A) 4 (B) 8
(C) 20 (D) 40

35. A shopkeeper has 84 kg of one type of rice and 108 kg of another. He packs each type into bags of equal, largest possible size (one type per bag). The bag size is:

- (A) 6 kg (B) 24 kg
(C) 12 kg (D) 4 kg

36. For the rice in the previous question, the total number of bags is:

- (A) 32 (B) 7
(C) 9 (D) 16

37. Which pair has its LCM equal to the larger of the two numbers?

- (A) 6 and 8 (B) 4 and 6
(C) 6 and 12 (D) 9 and 15

38. If a and b are two numbers with a a factor of b , then $\text{HCF}(a, b)$ and $\text{LCM}(a, b)$ are:

- (A) $\text{HCF} = b$, $\text{LCM} = a$ (B) $\text{HCF} = a$, $\text{LCM} = b$
(C) $\text{HCF} = 1$, $\text{LCM} = ab$ (D) $\text{HCF} = a$, $\text{LCM} = ab$

39. Two bells ring at intervals of 18 minutes and 24 minutes. If they ring together at 9:00 am, they next ring together at:

- (A) 9:42 am (B) 10:00 am
(C) 11:00 am (D) 10:12 am

40. The smallest number which when divided by 12, 15 and 20 leaves remainder 5 in each case is:

- (A) 60 (B) 65
(C) 305 (D) 55

41. The HCF of two consecutive whole numbers (such as 14 and 15) is always:

- (A) 1 (B) 2
(C) the smaller number (D) their product

42. The LCM of two numbers is 12 times their HCF. The sum of the HCF and LCM is 403, and one number is 93. The other number is:

- (A) 93 (B) 31
(C) 372 (D) 124

43. Find the smallest 4-digit number exactly divisible by 8, 12 and 18.

- (A) 1008 (B) 1080
(C) 1044 (D) 1000

44. A florist has 24 roses and 36 tulips. She makes identical bouquets using all the flowers, each bouquet with the same number of roses and the same number of tulips. The largest number of bouquets she can make is:

- (A) 6 (B) 24
(C) 12 (D) 4

45. For the bouquets in the previous question, each bouquet contains:

- (A) 4 roses and 6 tulips (B) 2 roses and 3 tulips
(C) 3 roses and 2 tulips (D) 6 roses and 9 tulips

46. The LCM of two numbers is 180 and their HCF is 6. If one number is 36, the other is:

- (A) 30 (B) 60
(C) 45 (D) 90

47. Which of these CANNOT be the HCF and LCM of a pair of numbers?

- (A) HCF 4, LCM 24 (B) HCF 6, LCM 90
(C) HCF 8, LCM 60 (D) HCF 5, LCM 60

48. Three pieces of wire of lengths 144 cm, 180 cm and 192 cm are cut into equal pieces of the greatest possible length. The number of such pieces is:

- (A) 12 (B) 43
(C) 36 (D) 16

49. Two numbers are co-prime. Which of these can be their LCM if the numbers are 12 and 35?

- (A) 7 (B) 105
(C) 60 (D) 420

50. A number when divided by 6, 9 and 15 leaves no remainder, and lies between 100 and 200. The number is:

- (A) 90 (B) 135
(C) 180 (D) 120

51. Mira jogs around a track in 4 minutes, Sam in 6 minutes, and Lina in 9 minutes. Starting together, they are all back at the start together after:

- (A) 18 min (B) 36 min
(C) 24 min (D) 54 min

52. If $\text{HCF}(x, 18) = 6$ and x is a two-digit number less than 30, then x could be:

- (A) 24 (B) 15
(C) 18 (D) 27

53. The product of the HCF and LCM of 16 and 20 is:

- (A) 4 (B) 80
(C) 320 (D) 36

54. The greatest number of children among whom 25 sweets and 35 chocolates can be distributed so that each child gets the same number of sweets and the same number of chocolates is:

- (A) 60 (B) 7
(C) 10 (D) 5

55. Two numbers have a product of 2400 and an LCM of 120. Their HCF is:

- (A) 10 (B) 20
(C) 24 (D) 12

56. The smallest number that, when 7 is added to it, becomes exactly divisible by 8, 12 and 16 is:

- | | |
|--------|--------|
| (A) 41 | (B) 48 |
| (C) 55 | (D) 89 |

57. Buses to two towns leave a stop every 15 minutes and every 25 minutes. They leave together at 8:00 am. The next time both leave together is:

- | | |
|-------------|-------------|
| (A) 8:40 am | (B) 9:30 am |
| (C) 9:00 am | (D) 9:15 am |

58. A rectangular courtyard 5 m 25 cm long and 3 m 78 cm broad is to be paved with identical square tiles of the largest possible size. The side of each tile is:

- | | |
|-----------|----------|
| (A) 12 cm | (B) 7 cm |
| (C) 21 cm | (D) 9 cm |

59. For the courtyard in the previous question, the least number of square tiles required is:

- | | |
|---------|---------|
| (A) 450 | (B) 525 |
| (C) 378 | (D) 900 |

60. The HCF of two numbers is 13 and their LCM is 1989. How many such pairs of numbers (a, b) with $a < b$ exist?

- | | |
|-------|-------|
| (A) 1 | (B) 4 |
| (C) 3 | (D) 2 |

End of question paper. Maximum marks: 240. Marking: +4 correct, -1 wrong, 0 unattempted.

Solutions — Mathematics (Class 7), Chapter 11: Finding Common Ground

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	D	21	A	31	C	41	A	51	B
2	C	12	C	22	B	32	D	42	D	52	A
3	A	13	B	23	C	33	B	43	A	53	C
4	C	14	C	24	D	34	A	44	C	54	D
5	D	15	A	25	A	35	C	45	B	55	B
6	A	16	D	26	B	36	D	46	A	56	A
7	C	17	A	27	D	37	C	47	C	57	D
8	D	18	B	28	B	38	B	48	B	58	C
9	A	19	C	29	C	39	D	49	D	59	A
10	B	20	D	30	A	40	B	50	C	60	D

Answer-key distribution: A = 15, B = 14, C = 16, D = 15.

Worked Solutions

1. (B) The HCF (Highest Common Factor) is the largest number that divides both numbers exactly. (A) describes the LCM — the smallest number both divide into. The HCF is about dividing INTO the numbers; the LCM is what the numbers divide into.

2. (C) $24 = 2^3 \times 3$, $36 = 2^2 \times 3^2$. Common primes to the lowest power: $2^2 \times 3 = 12$. (72 is the LCM — taking highest powers gives the trap.)

3. (A) $12 = 2^2 \times 3$, $18 = 2 \times 3^2$. LCM takes every prime to its highest power: $2^2 \times 3^2 = 36$. (6 is the HCF; $216 = 12 \times 18$ is the product, which only equals the LCM for co-prime numbers.)

4. (C) The tile side must divide both 12 and 16 exactly, so it must be a common factor; the largest is HCF(12, 16). $12 = 2^2 \times 3$, $16 = 2^4$, so $\text{HCF} = 2^2 = 4$ ft. (16 ft is the "HCF = the bigger number" trap; the HCF can't exceed the smaller dimension.)

5. (D) With 4 ft tiles the floor is $(12 \div 4) \times (16 \div 4) = 3 \times 4 = 12$ tiles. (A bigger tile gives FEWER tiles, so the HCF gives the fewest tiles for an exact fit.)

6. (A) $8 = 2^3$ and $15 = 3 \times 5$ share no prime, so $\text{HCF}(8, 15) = 1$ — co-prime. The others share a factor: 8 & 12 share 4, 9 & 15 share 3, 10 & 25 share 5. (Co-prime numbers need not themselves be prime — 8 and 15 are both composite.)

7. (C) 7 and 9 are co-prime ($\text{HCF} = 1$), so their LCM equals their product: $7 \times 9 = 63$.

8. (D) They ring together at common multiples of 6 and 8; the first is $\text{LCM}(6, 8)$. $6 = 2 \times 3$, $8 = 2^3$, $\text{LCM} = 2^3 \times 3 = 24$ minutes. ($48 = 6 \times 8$ is the trap — the product overshoots because they share the factor 2.)

9. (A) $\text{HCF} \times \text{LCM} = \text{product}$, so $\text{LCM} = \text{product} \div \text{HCF} = 216 \div 6 = 36$. ($1296 = 216 \times 6$ multiplies instead of divides; $210 = 216 - 6$ subtracts.)

10. (B) For any two numbers, $\text{HCF} \times \text{LCM} = \text{the product of the two numbers}$. This lets you find one if you know the other.

11. (D) HCF takes each COMMON prime to its LOWEST power: from $2^2 \times 3$ and 2×3^2 , the lowest powers are 2^1 and 3^1 , giving $2 \times 3 = 6$. (36 uses highest powers — that's the LCM; 12 keeps 2^2 instead of the lower 2^1 .)

12. (C) LCM takes every prime to its HIGHEST power: 2^2 (from 12) and 3^2 (from 18), giving $2^2 \times 3^2 = 36$. (6 uses lowest powers — that's the HCF; 216 multiplies the two numbers.)

13. (B) $\text{HCF} \times \text{LCM} = \text{product of the two numbers}$: $6 \times 72 = 432$. Other number $= 432 \div 24 = 18$. Check: $\text{HCF}(24, 18) = 6$ and $\text{LCM}(24, 18) = 72$. ✓

14. (C) Convert to cm: 700, 385, 1295. The longest exact measuring length is $\text{HCF}(700, 385, 1295)$. $700 = 2^2 \times 5^2 \times 7$, $385 = 5 \times 7 \times 11$, $1295 = 5 \times 7 \times 37$. Common: $5 \times 7 = 35$ cm. ✓

15. (A) The HCF divides the smaller number, so it can never exceed it — always true. (B) is false (the LCM is usually larger than both); (C) fails for e.g. 12 and 16 ($\text{HCF} 4$ is neither); (D) holds only for co-prime pairs.

16. (D) $18 = 2 \times 3^2$, $24 = 2^3 \times 3$, $30 = 2 \times 3 \times 5$. Common primes to lowest power: $2 \times 3 = 6$.

17. (A) $4 = 2^2$, $6 = 2 \times 3$, $8 = 2^3$. LCM takes highest powers: $2^3 \times 3 = 24$. Check: $24 \div 4$, $\div 6$, $\div 8$ are all whole. ✓

18. (B) All three change together at $\text{LCM}(24, 36, 54)$. $24 = 2^3 \times 3$, $36 = 2^2 \times 3^2$, $54 = 2 \times 3^3$. Highest powers: $2^3 \times 3^3 = 8 \times 27 = 216$ s. (6 s is the HCF — the trap of dividing instead.)

19. (C) "Divides both leaving remainder 0" means a common factor; the greatest is $\text{HCF}(70, 125)$. $70 = 2 \times 5 \times 7$, $125 = 5^3$. Common: 5.

- 20. (D)** Numbers in ratio 3 : 4 can be written $3k$ and $4k$. Since 3 and 4 are co-prime, their HCF is k . Given $\text{HCF} = 5$, take $k = 5$: the numbers are 15 and 20. Check: $\text{HCF}(15, 20) = 5$ and $15 : 20 = 3 : 4$. ✓ (30 and 40 have HCF 10, not 5; 3 and 4 have HCF 1.)
- 21. (A)** $\text{HCF} \times \text{LCM} = \text{product}$, so $\text{HCF} = (15 \times 20) \div \text{LCM} = 300 \div 60 = 5$. (300 forgets to divide.)
- 22. (B)** Common multiples of 4 and 6 are multiples of $\text{LCM}(4, 6) = 12$: 12, 24, 36, 48... 18 is a multiple of 6 but not of 4, so it is NOT a common multiple. (12, 24, 36 all are.)
- 23. (C)** Common factors of two numbers are exactly the factors of their HCF. $\text{HCF}(24, 36) = 12$, and 12 has factors 1, 2, 3, 4, 6, 12 — that is 6 common factors.
- 24. (D)** The gears realign after a number of teeth that is a multiple of both 8 and 12 — the LCM. $8 = 2^3$, $12 = 2^2 \times 3$, $\text{LCM} = 2^3 \times 3 = 24$ teeth. ($96 = 8 \times 12$ overshoots.)
- 25. (A)** $\text{HCF} = 1$ means the numbers are co-prime, and for co-prime numbers $\text{LCM} = \text{product}$. (B) is false — 8 and 15 are co-prime yet neither is prime; (C) and (D) are false (equal numbers share a factor > 1 , and if one divides the other the HCF is the smaller, not 1).
- 26. (B)** Euclid: $2346 = 1 \times 1794 + 552$; $1794 = 3 \times 552 + 138$; $552 = 4 \times 138 + 0$. So $\text{HCF} = 138$. ($138 = 2 \times 3 \times 23$.) Listing factors here would be hopeless — that's the point of Euclid's method.
- 27. (D)** Smallest number divisible by 1–6 = $\text{LCM}(1, 2, 3, 4, 5, 6)$. Highest powers among them: 2^2 (from 4), 3 (from 3 or 6), 5 (from 5): $4 \times 3 \times 5 = 60$. (120 and 720 are larger common multiples, not the LEAST.)
- 28. (B)** Greatest equal piece length = $\text{HCF}(36, 48)$. $36 = 2^2 \times 3^2$, $48 = 2^4 \times 3$, $\text{HCF} = 2^2 \times 3 = 12$ cm. (24 divides 48 but not 36.)
- 29. (C)** With 12 cm pieces: $36 \div 12 + 48 \div 12 = 3 + 4 = 7$ pieces.
- 30. (A)** If h divides both a and b , then it divides $a + b$ and $a - b$ (since each is a sum/difference of multiples of h). This is exactly why Euclid's algorithm works.
- 31. (C)** $360 = 2 \times 180 = 2^2 \times 90 = 2^3 \times 45 = 2^3 \times 9 \times 5 = 2^3 \times 3^2 \times 5$. Check: $8 \times 9 \times 5 = 360$. ✓ ($2^3 \times 3 \times 5 = 120$ and $2^2 \times 3^2 \times 5 = 180$ are wrong prime powers.)
- 32. (D)** HCF takes common primes to the lowest power. Both have 2 (powers 2 and 1 → take 2^1) and 3 (powers 1 and 2 → take 3^1); 5 and 7 are not common. $\text{HCF} = 2 \times 3 = 6$. (1260 is the LCM; 210 wrongly includes the non-common 5 and 7.)
- 33. (B)** LCM takes every prime to its highest power: 2^2 (from the first), 3^2 (from the second), and 5 and 7 once each: $2^2 \times 3^2 \times 5 \times 7 = 4 \times 9 \times 35 = 1260$. (6 is the HCF.)

34. (A) A number leaving the same remainder when dividing 43, 91, 183 must divide their differences exactly: $91 - 43 = 48$, $183 - 91 = 92$, $183 - 43 = 140$. $\text{HCF}(48, 92, 140)$: $48 = 2^4 \times 3$, $92 = 2^2 \times 23$, $140 = 2^2 \times 5 \times 7 \rightarrow$ common $2^2 = 4$. Check: $43 = 4 \times 10 + 3$, $91 = 4 \times 22 + 3$, $183 = 4 \times 45 + 3$. ✓

35. (C) Largest equal bag size (one type per bag) = $\text{HCF}(84, 108)$. $84 = 2^2 \times 3 \times 7$, $108 = 2^2 \times 3^3$, $\text{HCF} = 2^2 \times 3 = 12$ kg. (24 divides neither exactly.)

36. (D) With 12 kg bags: $84 \div 12 + 108 \div 12 = 7 + 9 = 16$ bags.

37. (C) The LCM equals the larger number exactly when the smaller number divides the larger. Testing each: 6 & 8 $\rightarrow$ LCM 24; 4 & 6 $\rightarrow$ LCM 12; 9 & 15 $\rightarrow$ LCM 45; but 6 & 12 $\rightarrow$ 6 divides 12, so LCM = 12 = the larger number. Hence (C) 6 and 12.

38. (B) If a is a factor of b , then a divides b , so the largest common factor is a itself ($\text{HCF} = a$) and the smallest common multiple is b itself ($\text{LCM} = b$). E.g. $a = 4$, $b = 12$: $\text{HCF} = 4$, $\text{LCM} = 12$.

39. (D) Next together after $\text{LCM}(18, 24)$ minutes. $18 = 2 \times 3^2$, $24 = 2^3 \times 3$, $\text{LCM} = 2^3 \times 3^2 = 72$ min = 1 h 12 min. $9:00 + 1:12 = 10:12$ am. (9:42 adds only 42 min; 11:00 overshoots.)

40. (B) Remainder 5 each time means (number $- 5$) is divisible by 12, 15, 20 $\rightarrow$ it is a multiple of $\text{LCM}(12, 15, 20)$. $12 = 2^2 \times 3$, $15 = 3 \times 5$, $20 = 2^2 \times 5 \rightarrow \text{LCM} = 2^2 \times 3 \times 5 = 60$. Smallest number = $60 + 5 = 65$. (60 forgets to add the remainder.)

41. (A) Two consecutive whole numbers share no common factor greater than 1 (any common divisor would divide their difference 1), so their HCF is always 1 — they are always co-prime.

42. (D) Let $\text{HCF} = h$; $\text{LCM} = 12h$. $h + 12h = 403 \rightarrow 13h = 403 \rightarrow h = 31$, $\text{LCM} = 372$. $\text{HCF} \times \text{LCM} = \text{product} = 31 \times 372 = 11532$. Other number = $11532 \div 93 = 124$. Check: $\text{HCF}(93, 124) = 31$, $\text{LCM} = 372$. ✓

43. (A) Must be divisible by $\text{LCM}(8, 12, 18)$. $8 = 2^3$, $12 = 2^2 \times 3$, $18 = 2 \times 3^2 \rightarrow \text{LCM} = 2^3 \times 3^2 = 72$. Smallest 4-digit multiple of 72: $1000 \div 72 \approx 13.9$, so $14 \times 72 = 1008$. ✓ (1080 = 15×72 is larger; 1000 isn't a multiple of 72.)

44. (C) Each bouquet gets an equal share of both, so the number of bouquets must divide 24 and 36; the largest is $\text{HCF}(24, 36) = 12$. ✓

45. (B) With 12 bouquets: $24 \div 12 = 2$ roses and $36 \div 12 = 3$ tulips each.

46. (A) $\text{HCF} \times \text{LCM} = \text{product}$: $6 \times 180 = 1080$. Other number = $1080 \div 36 = 30$. Check: $\text{HCF}(36, 30) = 6$, $\text{LCM} = 180$. ✓

47. (C) The HCF must divide the LCM for a valid pair. (A) $4 \mid 24$ ✓; (B) $6 \mid 90$ ✓; (D) $5 \mid 60$ ✓; (C) $8 \nmid 60$ ($60 \div 8$ isn't whole) — impossible. So (C) cannot occur.

48. (B) Greatest equal piece length = $\text{HCF}(144, 180, 192)$. $144 = 2^4 \times 3^2$, $180 = 2^2 \times 3^2 \times 5$, $192 = 2^6 \times 3 \rightarrow$ common $2^2 \times 3 = 12$ cm. Pieces = $144/12 + 180/12 + 192/12 = 12 + 15 + 16 = 43$.

49. (D) 12 and 35 are co-prime ($12 = 2^2 \times 3$, $35 = 5 \times 7$ share nothing), so $\text{LCM} = \text{product} = 12 \times 35 = 420$. (7 is just a factor; $105 = 3 \times 5 \times 7$ misses the 2^2 ; 60 ignores 35.)

50. (C) Divisible by 6, 9 and 15 means a multiple of $\text{LCM}(6, 9, 15)$. $6 = 2 \times 3$, $9 = 3^2$, $15 = 3 \times 5 \rightarrow \text{LCM} = 2 \times 3^2 \times 5 = 90$. Multiples of 90 between 100 and 200: 180. (90 is below 100; 135 is divisible by 9 and 15 but not 6.)

51. (B) All back together after $\text{LCM}(4, 6, 9)$. $4 = 2^2$, $6 = 2 \times 3$, $9 = 3^2 \rightarrow \text{LCM} = 2^2 \times 3^2 = 36$ min. (18 is a multiple of 6 and 9 but not 4.)

52. (A) Test HCF with $18 = 2 \times 3^2$ for each option. $24 = 2^3 \times 3 \rightarrow$ common primes to lowest power $2^1 \times 3^1 = 6$ ✓. The others fail: $\text{HCF}(15, 18) = 3$, $\text{HCF}(18, 18) = 18$, $\text{HCF}(27, 18) = 9$. Only 24 gives HCF 6.

53. (C) Product of HCF and LCM equals the product of the numbers: $16 \times 20 = 320$. (No need to find HCF and LCM separately — the identity gives it directly. $\text{HCF}(16, 20) = 4$, $\text{LCM} = 80$, and $4 \times 80 = 320$. ✓)

54. (D) Each child gets equal sweets and equal chocolates, so the number of children divides both 25 and 35; the greatest is $\text{HCF}(25, 35) = 5$. (Each child then gets 5 sweets and 7 chocolates.)

55. (B) $\text{HCF} = \text{product} \div \text{LCM} = 2400 \div 120 = 20$. Check: HCF must divide LCM, and $20 \mid 120$. ✓

56. (A) number + 7 is divisible by 8, 12, 16 $\rightarrow$ a multiple of $\text{LCM}(8, 12, 16) = 2^4 \times 3 = 48$. Smallest such: 48, so number = $48 - 7 = 41$. Check: $41 + 7 = 48 \div 8, \div 12, \div 16$ all whole. ✓

57. (D) Leave together at common multiples of 15 and 25 $\rightarrow \text{LCM}(15, 25)$. $15 = 3 \times 5$, $25 = 5^2 \rightarrow \text{LCM} = 3 \times 5^2 = 75$ min = 1 h 15 min. 8:00 + 1:15 = 9:15 am. (8:40 adds only 40 min; 9:30 = 90 min overshoots.)

58. (C) Convert: 525 cm $\times$ 378 cm. Largest square tile side = $\text{HCF}(525, 378)$. $525 = 3 \times 5^2 \times 7$, $378 = 2 \times 3^3 \times 7 \rightarrow$ common $3 \times 7 = 21$ cm. (12, 9 don't divide 525.)

59. (A) With 21 cm tiles: $(525 \div 21) \times (378 \div 21) = 25 \times 18 = 450$ tiles.

60. (D) $\text{HCF} = 13$ means both numbers are $13a$ and $13b$ with a, b co-prime; $\text{LCM} = 13ab = 1989 \rightarrow ab = 153 = 9 \times 17 = 3^2 \times 17$. Co-prime factor pairs (a, b) of 153 with $a < b$: (1, 153)

and (9, 17). That gives 2 pairs: (13, 1989) and (117, 221). So 2 such pairs. ✓

Solutions complete: 60 / 60.